Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 36

Pascal's Triangle and the Binomial Theorem

Covers Chartrand & Zhang, Section 9.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

The three facts

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \qquad \binom{n}{r} = \binom{n}{n-r} \qquad (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r$$

For coefficient problems: write the general term first

For $(ax+by)^n$ the general term is $\binom{n}{r}(ax)^{n-r}(by)^r$. Expand the constants, match the exponent you need, solve for r, then evaluate. **Do not guess which binomial coefficient to use.**

Three mistakes to expect

1. **Forgetting the constants.** In $(3x+2y)^5$ the 3 and 2 carry powers too.
2. **Dropping the sign** in $(x-y)^n$, where the term carries $(-1)^r$.
3. **Solving for the wrong r.** Match the exponent carefully.

Part II: Practice Problems

Section A — The triangle

1.

(a) Write rows 7 and 8 of Pascal's triangle.

(b) Verify Theorem 9.1 for $\binom{8}{4}$.

(c) Verify that row 7 sums to 2^7.

2. Verify the Hockey Stick Theorem in the case $\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \binom{6}{3} = \binom{7}{4}$.

Section B — Combinatorial proof

3. Prove that $\binom{n}{r} = \binom{n}{n-r}$ by a counting argument, not by algebra.

Section C — The Binomial Theorem

4. Find each coefficient. Write the general term first.

(a) The coefficient of x^5y^3 in $(x+y)^8$.

(b) The coefficient of x^3y^2 in $(3x+2y)^5$.

(c) The coefficient of a^2b^5 in $(a-b)^7$.

5. Find the constant term in $\left(x + \dfrac{2}{x}\right)^6$.

Section D — Writing

6. In five to eight complete sentences, explain why the coefficients in $(x+y)^n$ are the same numbers that count subsets. Describe what is being chosen when you expand the product.

Section E — Challenge (optional)

7. Use the Binomial Theorem to prove that $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$, and then prove the same identity by a counting argument.